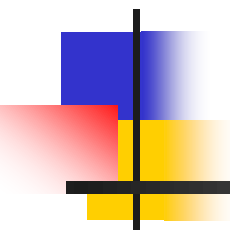
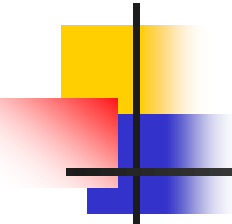


CHAPTER 3:

FREQUENCY-DOMAIN REPRESENTATION

OF SIGNALS

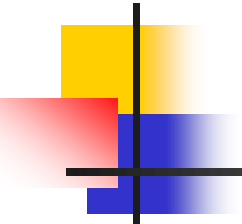
A decorative graphic is located on the left side of the slide. It features a black crosshair with a blue square in the top-left quadrant, a red square in the bottom-left quadrant, and a yellow square in the bottom-right quadrant. A horizontal grey line extends from the crosshair across the slide.



Why frequency-domain approach?

- Let us know other characteristics of signals, which are not exhibited on time-domain representation
 - E.g.: some voiced sounds are hardly distinguishable if only time-domain features are used
- Let us know other characteristics of systems, which are not exhibited on time-domain representation
 - Response of a LTI system to a sinusoid is a sinusoid with same frequency but different amplitude and phase
 - Details are presented in next chapter

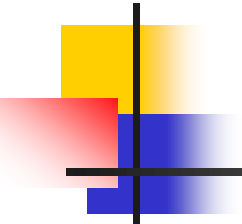
miền tần số cho thấy sự khác biệt giữa các tín hiệu, hệ thống mà miền thời gian không biểu diễn được



Basic ideas

- Any signal can be described as a sum or integral of sinusoidal signals tín hiệu bất kì bằng tổng của các tín hiệu sin
 - Exact form of the representation depends on whether
 - signal is continuous-time or discrete-time liên tục/ rời rạc
 - signal is periodic or aperiodic tuần hoàn / không tuần hoàn
- Clear understanding of sinusoidal signals are essential

4 công thức biểu diễn Fourier



Fourier representation of signals

■ Outline:

1. Sinusoidal signals and their properties
2. Summary of Fourier series and Fourier transforms

Continuous-time sinusoids

Definition

$$x(t) = A \cos(2\pi F_0 t + \theta), \quad -\infty < t < \infty$$

A: amplitude

θ : phase (radians)

F_0 : frequency (Hz)

angular frequency (rad/s)

$$\Omega_0 = 2\pi F_0$$

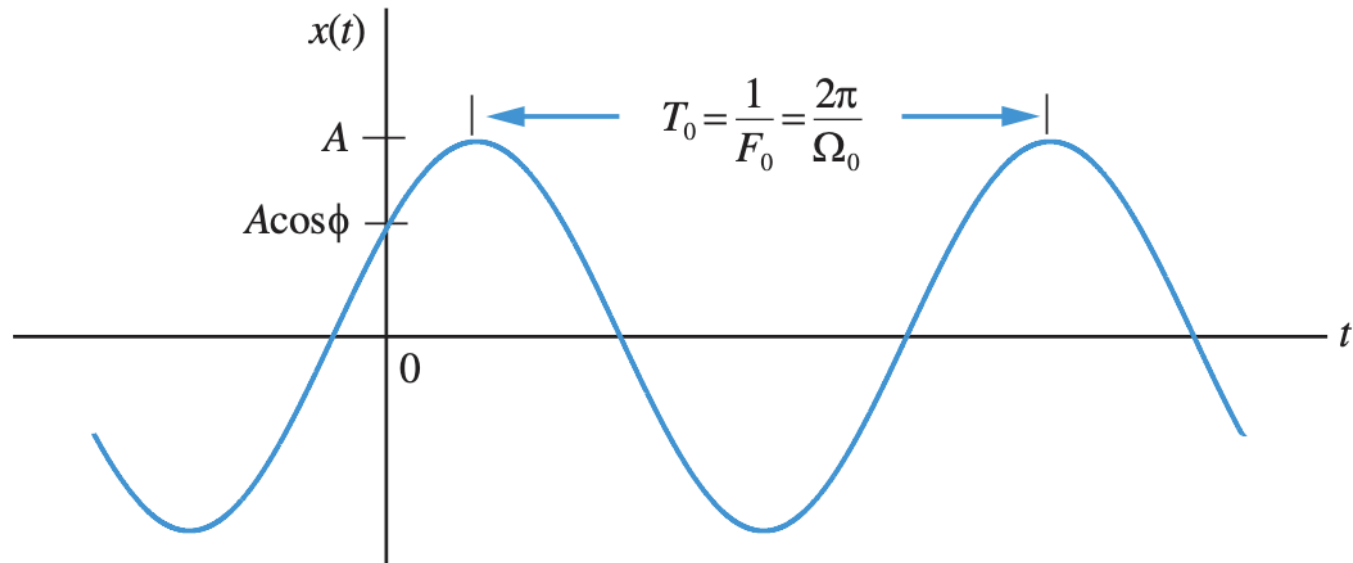
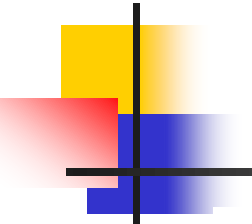


Figure 4.1 Continuous-time sinusoidal signal and its parameters.



Continuous-time sinusoids

Relation with complex exponentials

$$A \cos(\Omega_0 t + \theta) = \frac{A}{2} e^{j\theta} e^{j\Omega_0 t} + \frac{A}{2} e^{-j\theta} e^{-j\Omega_0 t}.$$

- Every sinusoidal signal is a sum of two complex exponentials with the same frequency
- Negative frequencies for mathematical convenience

Continuous-time sinusoids

Properties

- Two sinusoids with different frequencies are different
- Rate of oscillation increases indefinitely w/ increasing frequency

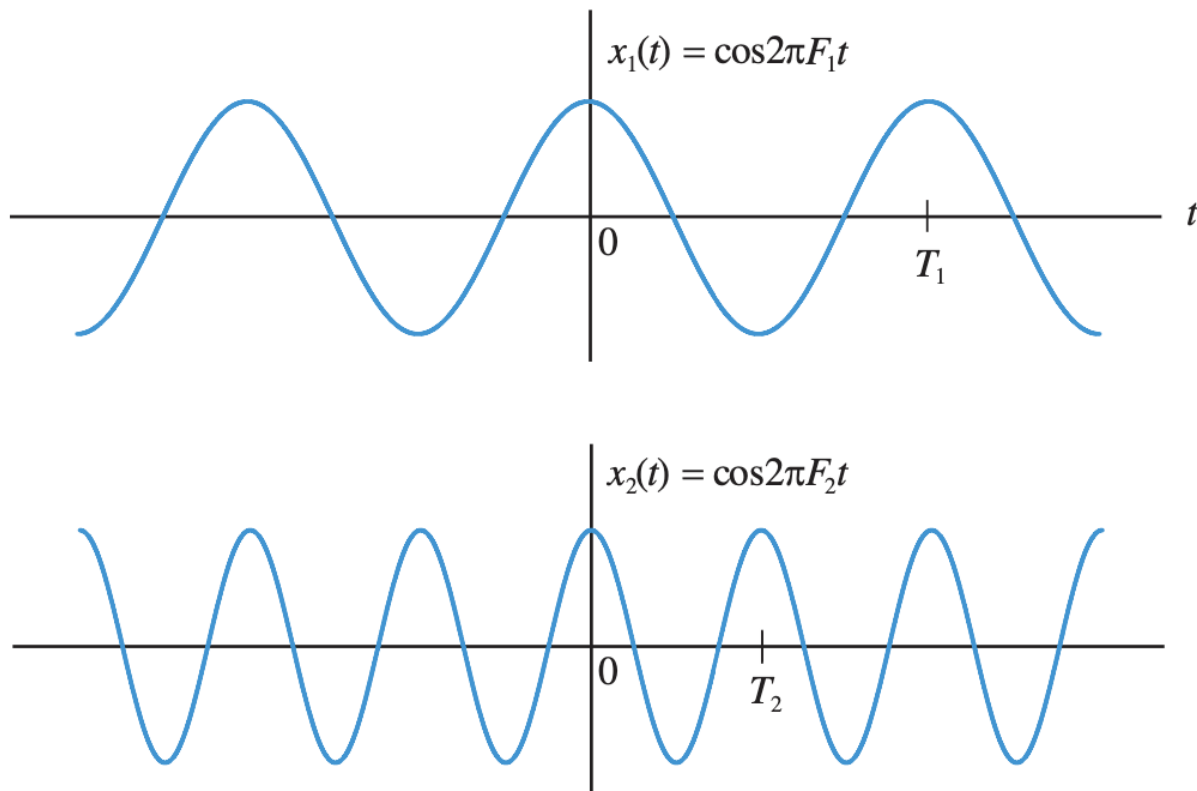


Figure 4.2 For continuous-time sinusoids, $F_1 < F_2$ always implies that $T_1 > T_2$.

Continuous-time sinusoids

Harmonically related complex exponentials

$$s_k(t) = e^{jk\Omega_0 t} = e^{j2\pi kF_0 t}, \quad k = 0, \pm 1, \pm 2, \dots$$

tín hiệu sin phức mang t/s góc kF_0 (tần số thực KF_0)

tần số cơ bản chỉ gắn với tín hiệu tuần hoàn

➤ Fundamental frequency: $\Omega_0 = 2\pi/T_0 = 2\pi F_0$

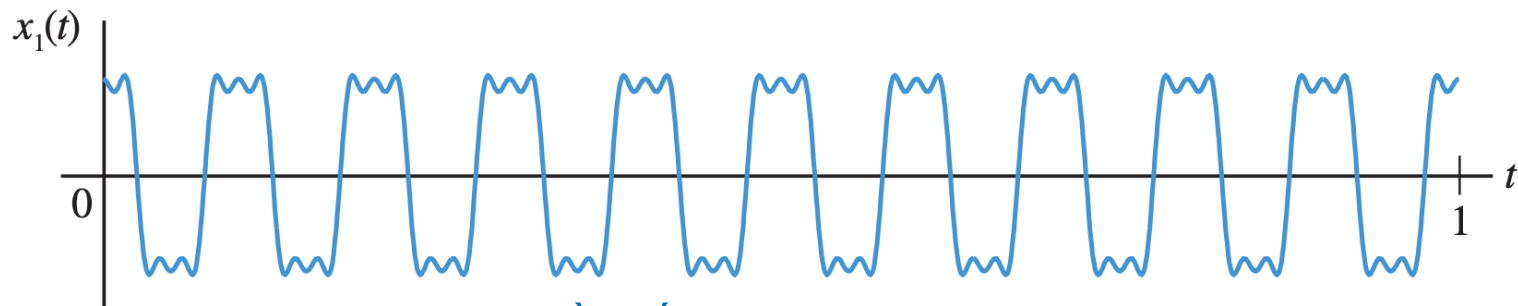
➤ $s_k(t)$ is the k th harmonic (“hài bậc k ”)

➤ Example of a periodic signal composed of three sinusoids with harmonically related frequencies

$$x_1(t) = \frac{1}{3} \cos(2\pi F_0 t) - \frac{1}{10} \cos(2\pi 3F_0 t) + \frac{1}{20} \cos(2\pi 5F_0 t)$$

trái sang phải : pt phân tích

phải sang trái: phương trình tổng hợp

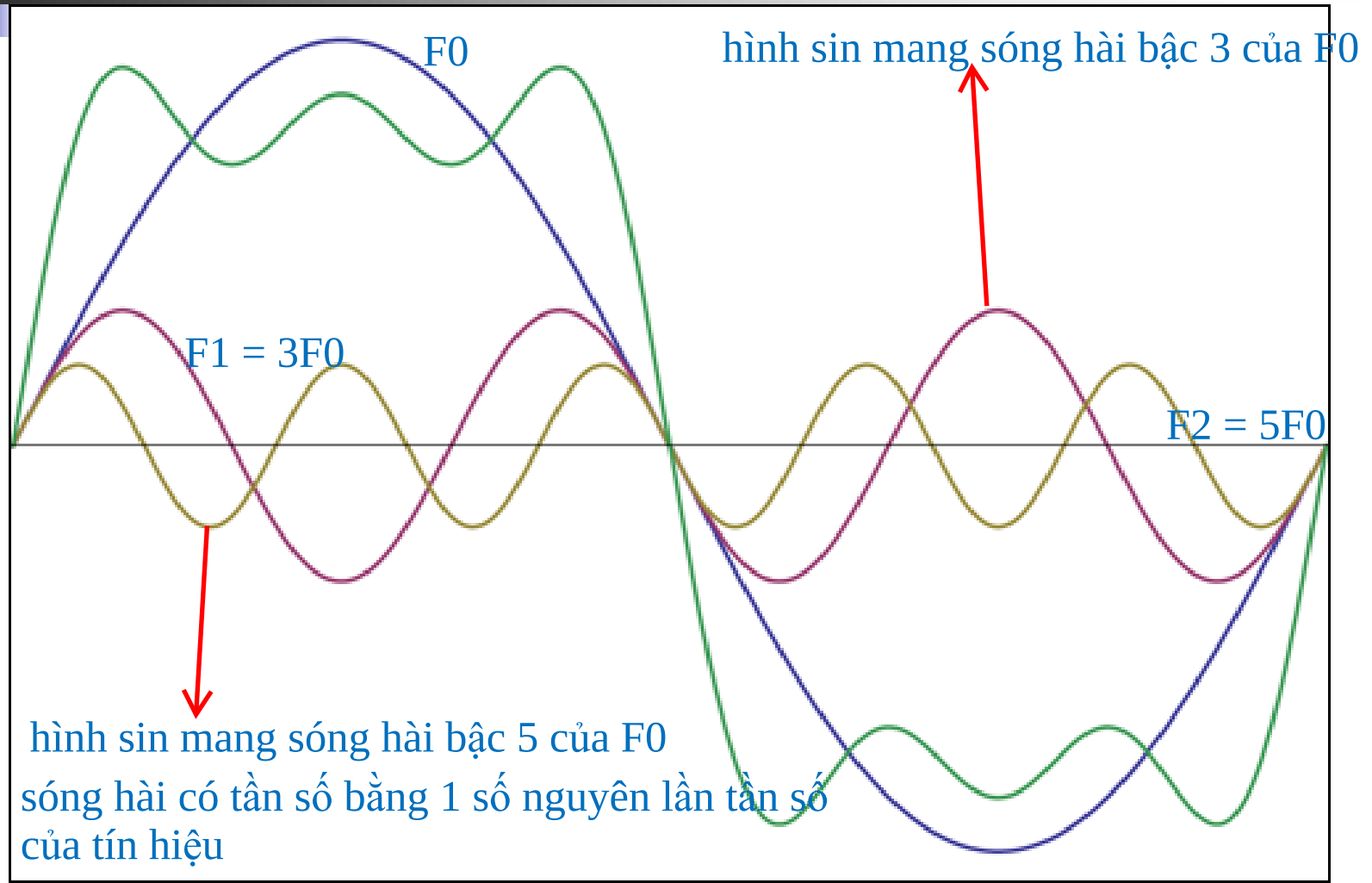


To: chu kỳ cơ bản

$F_0 = 1/T_0$: tần số cơ bản của $x_1(t)$

Continuous-time sinusoids

Harmonically related complex exponentials



(source: <https://electronics.stackexchange.com/questions/32310/what-exactly-are-harmonics-and-how-do-they-appear>)

Discrete-time sinusoids

Definition

Often obtained by sampling the continuous- time sinusoid

$$x[n] = x(nT) = A \cos(2\pi F_0 nT + \theta) = A \cos\left(2\pi \frac{F_0}{F_s} n + \theta\right)$$

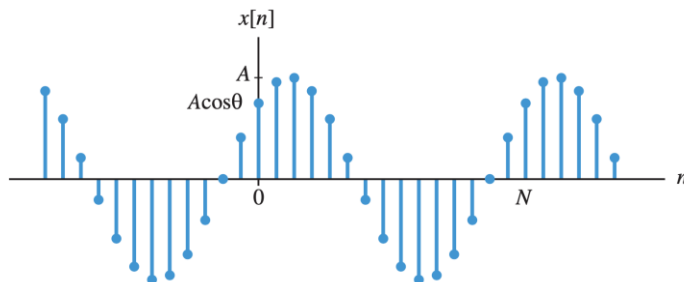
or $x[n] = A \cos(2\pi f_0 n + \theta) = A \cos(\omega_0 n + \theta)$

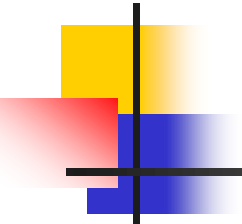
➤ Normalized frequency:

$$f \triangleq \frac{F}{F_s} = FT$$

➤ Normalized angular frequency:

$$\omega \triangleq 2\pi f = 2\pi \frac{F}{F_s} = \Omega T$$





Discrete-time sinusoids

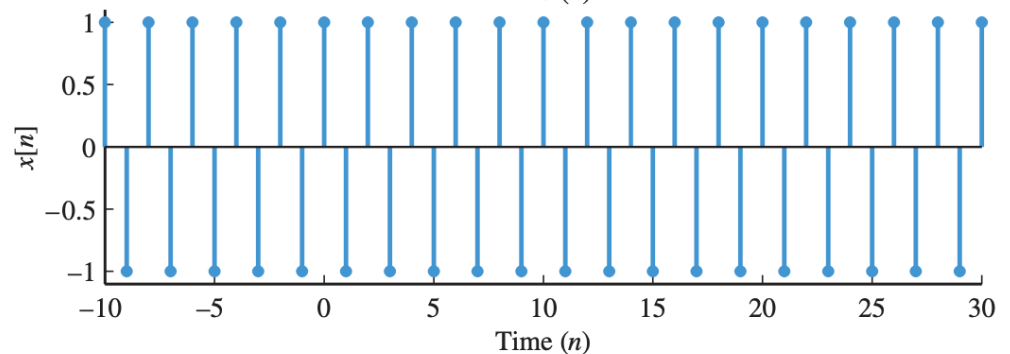
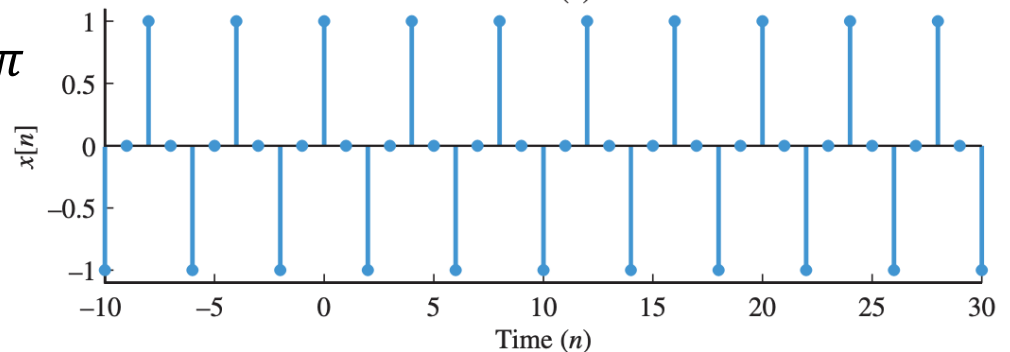
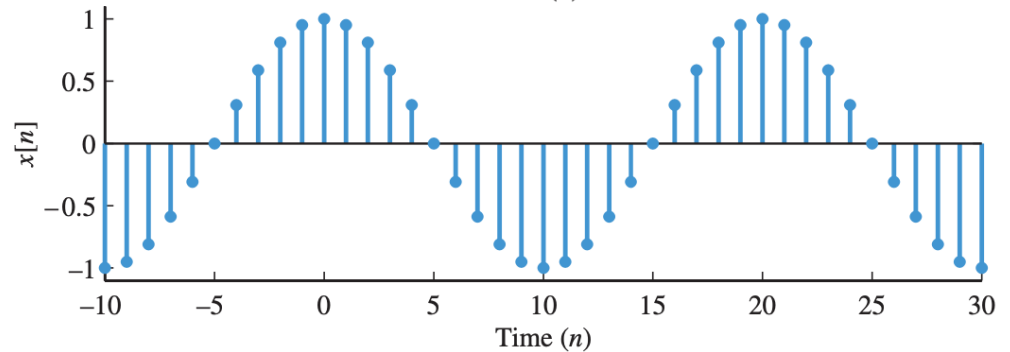
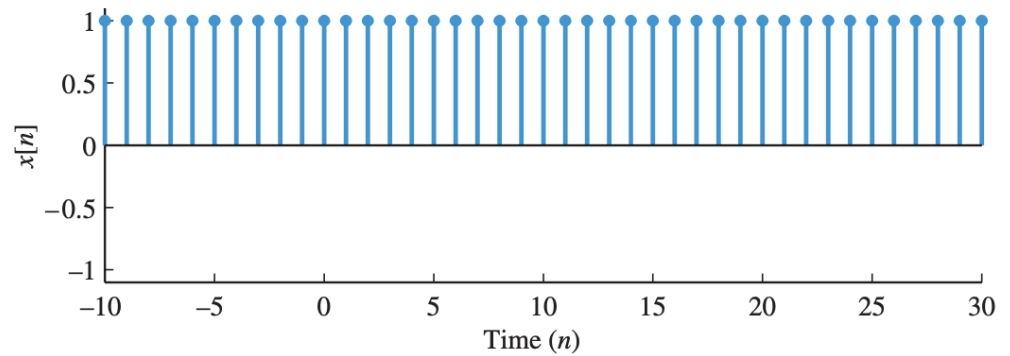
Properties

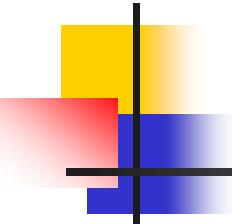
- $x[n] = A \cos(2\pi f_0 n + \theta)$ is periodic if and only if $f_0 = k/N$ (f_0 is a rational number)
 - If k and N are a pair of prime numbers, then N is the *fundamental period* of $x[n]$
- Sinusoidal sequences with angular frequencies separated by integer multiples of 2π are identical (*Periodicity in frequency*)
- All distinct sinusoidal sequences have frequencies within an interval of 2π radians
 - interested frequency ranges: $-\pi < \omega \leq \pi$ or $0 \leq \omega < 2\pi$

DT sinusoids

Properties

- Low frequencies (slow oscillations) at $\sim \omega_0 = k2\pi$
- High frequencies (rapid oscillations) at $\sim \omega_0 = \pi + k2\pi$





Discrete-time sinusoids

Harmonically related complex exponentials

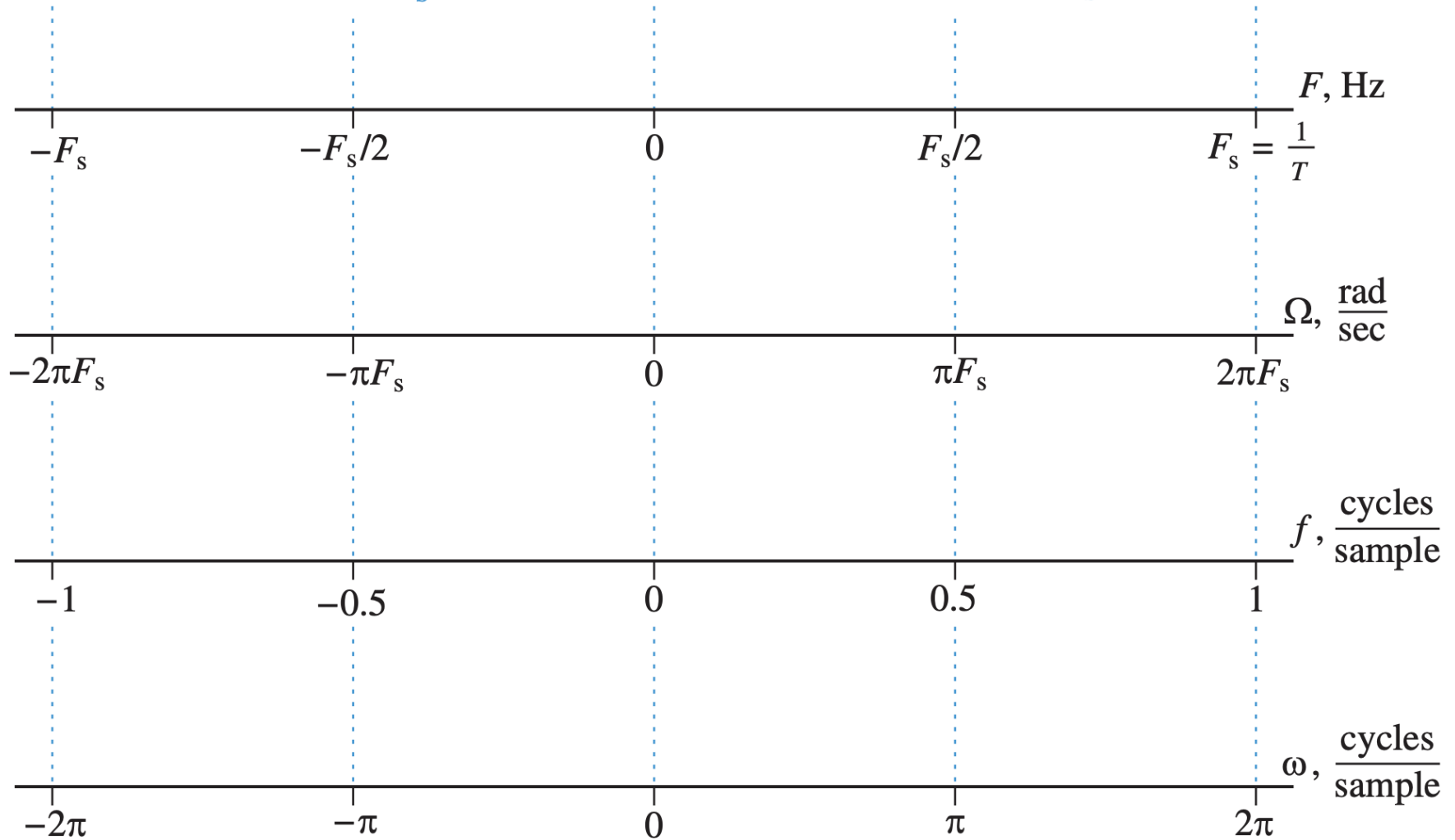
$$s_k[n] = A_k e^{j\omega_k n}. \quad -\infty < n < \infty$$

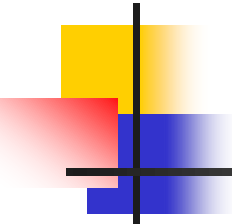
- For $s_k[n]$ to be periodic with fundamental period N , the frequency ω_k should be a rational multiple of 2π , that is, $\omega_k = 2\pi k/N$
- Fundamental frequency: $f_0 = 1/N$
- There are only N distinct harmonics at frequencies $f_k = k/N$ ($0 \leq k \leq N-1$)

Frequency variables and units

$$f \triangleq \frac{F}{F_s} = FT$$

$$\omega \triangleq 2\pi f = 2\pi \frac{F}{F_s} = \Omega T$$



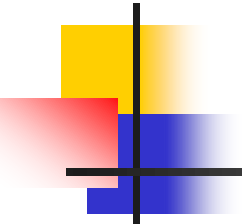
A decorative graphic in the top-left corner consisting of overlapping yellow, red, and blue squares with a black crosshair.

Fourier representation of signals

■ Outline:

1. Sinusoidal signals and their properties

2. Summary of Fourier series and Fourier transforms



Basic ideas revisited

- Any signal can be described as a sum or integral of sinusoidal signals
- Exact form of the representation depends on whether
 - signal is continuous-time (CT) or discrete-time (DT)
 - signal is periodic or aperiodic

Fourier series for CT periodic signals

- Continuous-Time Fourier Series (CTFS) pair

pt tổng hợp

pt phân tích

Fourier Synthesis Equation

Fourier Analysis Equation

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\Omega_0 t}$$

có c_k tìm $x(t)$

CTFS

$$c_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\Omega_0 t} dt.$$

có $x(t)$ tìm $c(k)$

- *Analysis equation*: analyzes (“breaks-up”) a periodic signal $x(t)$ into a set of harmonic components $\{c_k \exp(jk\Omega_0 t)\}$
- *Synthesis equation*: synthesizes the signal $x(t)$ from its harmonic components
 tạo nên/sinh ra/ tổng hợp

Fourier series for CT periodic signals

- phổ
■ Spectrum of CT periodic signals
 - các tần số phổ của tín hiệu tuần hoàn $x(t)$ ↑
 - $\{c_k\}$: Fourier series coefficients ($k = 0, \pm 1, \pm 2, \dots$)
 - *Spectrum*: the plot of c_k as a function of frequency $F = kF_0$, which constitutes a description of the signal in the frequency-domain

$$c_k = |c_k|e^{j\angle c_k}$$

ck là 1 chuỗi rời rạc theo chỉ số k

cho biến biên độ của sóng hài bậc k

$|c_k|$: magnitude spectrum of $x(t)$

phổ biên độ

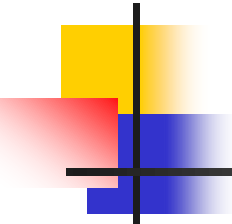
k: là biến tần số rời rạc
vì $F = kF_0$

$\angle c_k$: phase spectrum of $x(t)$

phổ pha: cho biến pha của sóng
hài bậc k

phổ: là 1 hàm theo biến tần số cho biết một tín hiệu gồm các tần số nào (mỗi tần số này có biên độ và pha bằng bao nhiêu), biểu diễn tín hiệu trên miền tần số

$F = kF_0 \Rightarrow$ phổ rời rạc (phổ vạch)



Fourier series for CT periodic signals

còn gọi là th công suất (vì cs < dương vô cùng)

■ Spectrum of CT periodic signals (cont.)

➤ $\{c_k\}$: Fourier series coefficients ($k = 0, \pm 1, \pm 2, \dots$)

công suất phân phối
liên tục theo thời gian
và rời rạc theo tần số

➤ *Parseval's relation*:

$$P_{av} = \frac{1}{T_0} \int_{T_0} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |c_k|^2$$

→ $|c_k|^2$: the portion of the average power of signal $x(t)$ that is contributed by the k th harmonic component

➤ *Power spectrum*: the plot of $|c_k|^2$ as a function of frequency $F = kF_0$

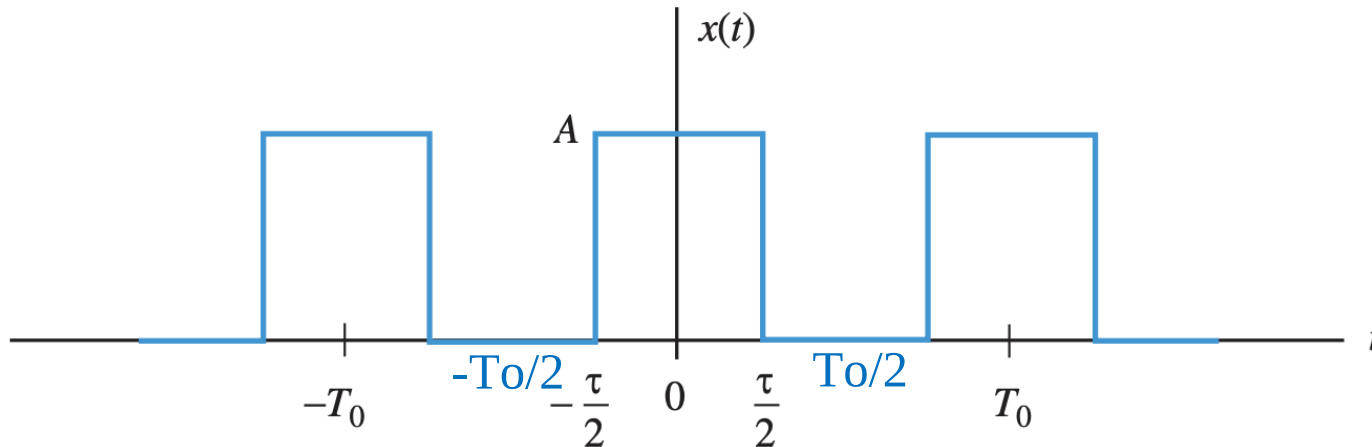
→ the power is distributed at a set of discrete frequencies

→ *discrete or line spectra* with uniform spacing $F_0 = 1/T_0$

(F_0 : fundamental frequency of $x(t)$)

Fourier series for CT periodic signals

- Example: Spectrum of rectangular pulse train



$$\begin{aligned} c_k &= \frac{1}{T_0} \int_{-\tau/2}^{\tau/2} A e^{-j2\pi k F_0 t} dt = \frac{A}{T_0} \left[\frac{e^{-j2\pi k F_0 t}}{-j2\pi k F_0} \right]_{-\tau/2}^{\tau/2} \\ &= \frac{A}{\pi F_0 k T_0} \frac{e^{j\pi k F_0 \tau} - e^{-j\pi k F_0 \tau}}{2j} \\ &= \frac{A\tau}{T_0} \frac{\sin \pi k F_0 \tau}{\pi k F_0 \tau}, \quad k = 0, \pm 1, \pm 2, \dots \end{aligned}$$

$$(c_0 = A\tau/T_0)$$

0,5đ : phổ gồm các vạch cách đều
nhau 1 khoảng $F_0 = 0.2 \text{ Hz}$

0.5đ: phổ gồm vô số thành phần hài

Fourier series for CT periodic signals

- Example: Spectrum of rectangular pulse train (cont.)

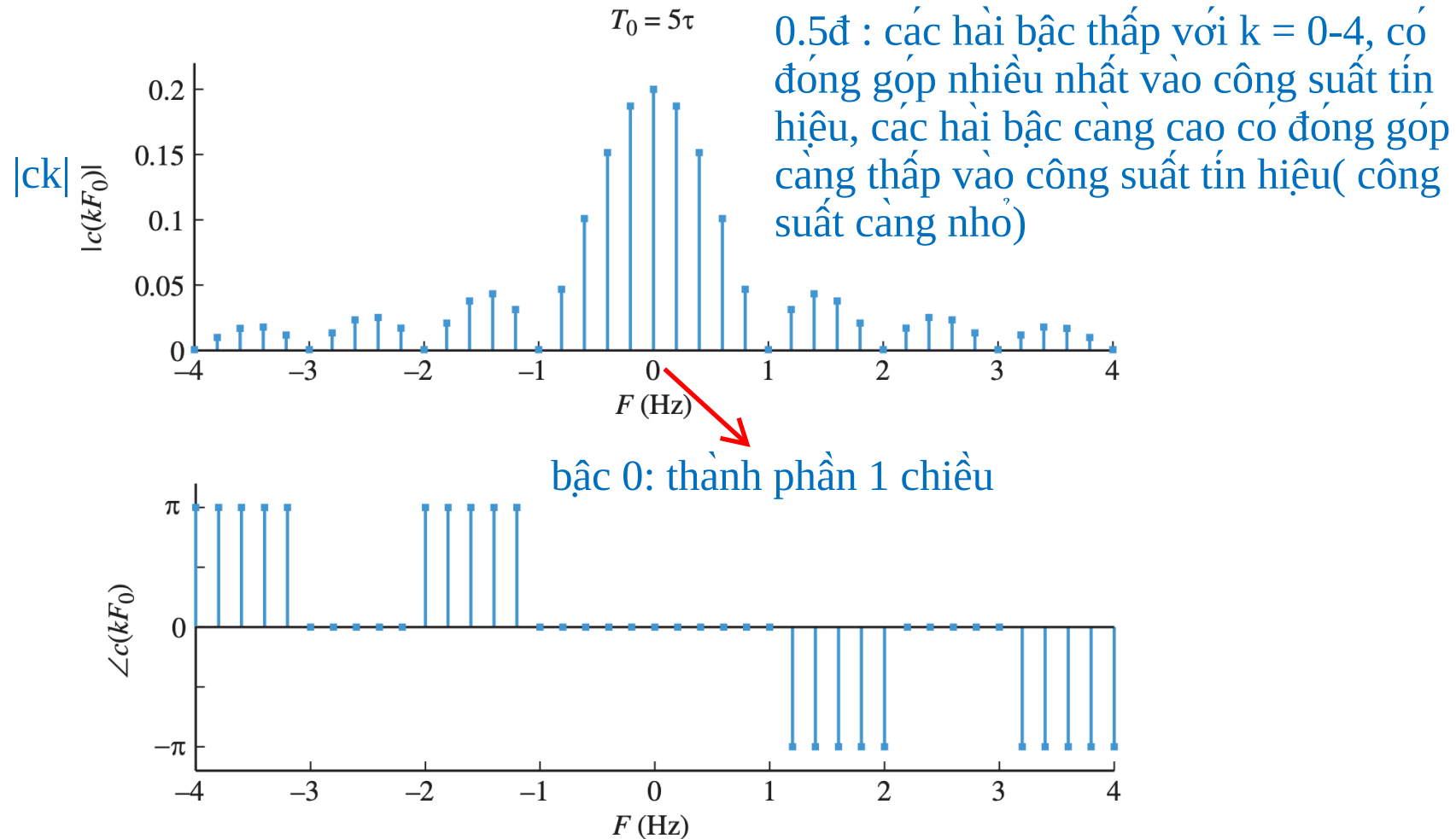


Figure 4.9 Magnitude and phase spectra of a rectangular pulse train with $A = 1$ and $T_0 = 5\tau = 5(\text{s})$

Fourier Transform for CT aperiodic signals

- Continuous-Time Fourier Transform (CTFT) pair

Fourier Synthesis Equation

Fourier Analysis Equation

$$x(t) = \int_{-\infty}^{\infty} X(j2\pi F) e^{j2\pi Ft} dF \xleftrightarrow{\text{CTFT}} X(j2\pi F) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi Ft} dt$$

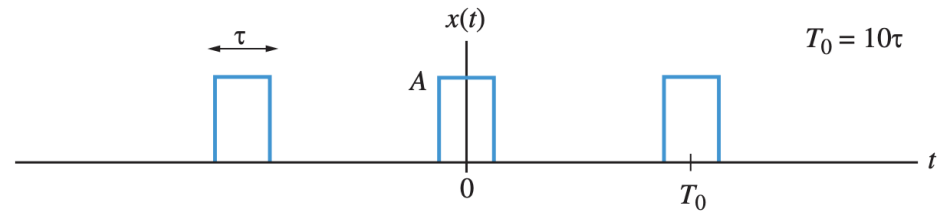
or

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega) e^{j\Omega t} d\Omega \xleftrightarrow{\text{CTFT}} X(j\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$$

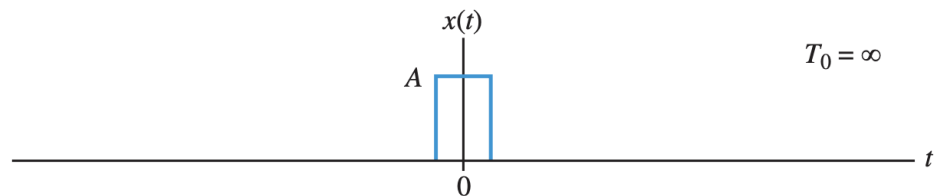
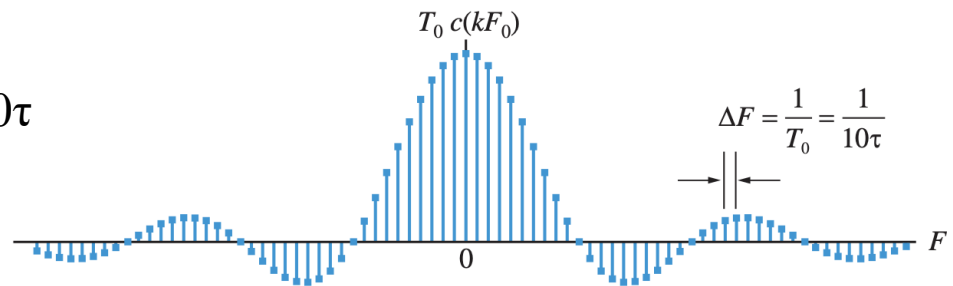
- $X(j2\pi F)$: *spectrum* of the aperiodic signal $x(t)$ (called Fourier Transform)
- CTFT is of the same nature as CTFS with fund. frequency $F_0 = 1/T_0 \rightarrow 0$

Fourier Transform for CT aperiodic signals

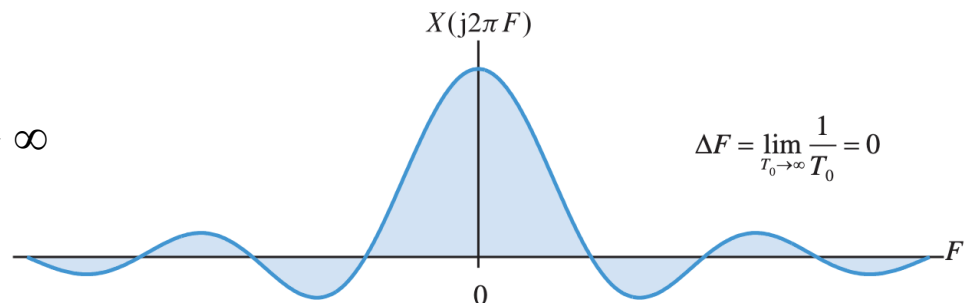
- Transition from the CTFS to CTFT



CTFS of periodic signal $x(t)$ when $T_0 = 10\tau$



CTFT of aperiodic signal $x(t)$ when $T_0 \rightarrow \infty$



Fourier Transform for CT aperiodic signals

- Parseval's relation for aperiodic signals with finite energy

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(j2\pi F)|^2 dF$$

→ total energy of $x(t)$ may be obtained either from the signal itself or from its spectrum

- $|X(j2\pi F)|^2 \Delta F$, for a small ΔF , provides the energy of the signal in a narrow frequency band of width ΔF
- *Energy-density spectrum*: the plot of $|X(j2\pi F)|^2$

Fourier Transform for CT aperiodic signals

- Example 1: Spectrum of causal exponential signal

$$x(t) = \begin{cases} e^{-at}, & t > 0 \\ 0, & t < 0 \end{cases}$$

This signal is absolutely integrable if $a > 0$.

$$X(j2\pi F) = \int_0^{\infty} e^{-at} e^{-j2\pi Ft} dt = -\frac{1}{a + j2\pi F} e^{-(a+j2\pi F)t} \bigg|_0^{\infty}.$$

Hence,

$$X(j2\pi F) = \frac{1}{a + j2\pi F} \quad \text{or} \quad X(j\Omega) = \frac{1}{a + j\Omega}, \quad a > 0$$

Magnitude and phase spectra:

$$|X(j2\pi F)| = \frac{1}{\sqrt{a^2 + (2\pi F)^2}}, \quad \angle X(j2\pi F) = -\tan^{-1} \left(2\pi \frac{F}{a} \right), \quad a > 0$$

Fourier Transform for CT aperiodic signals

- Example 1: Spectrum of causal exponential signal (cont.)

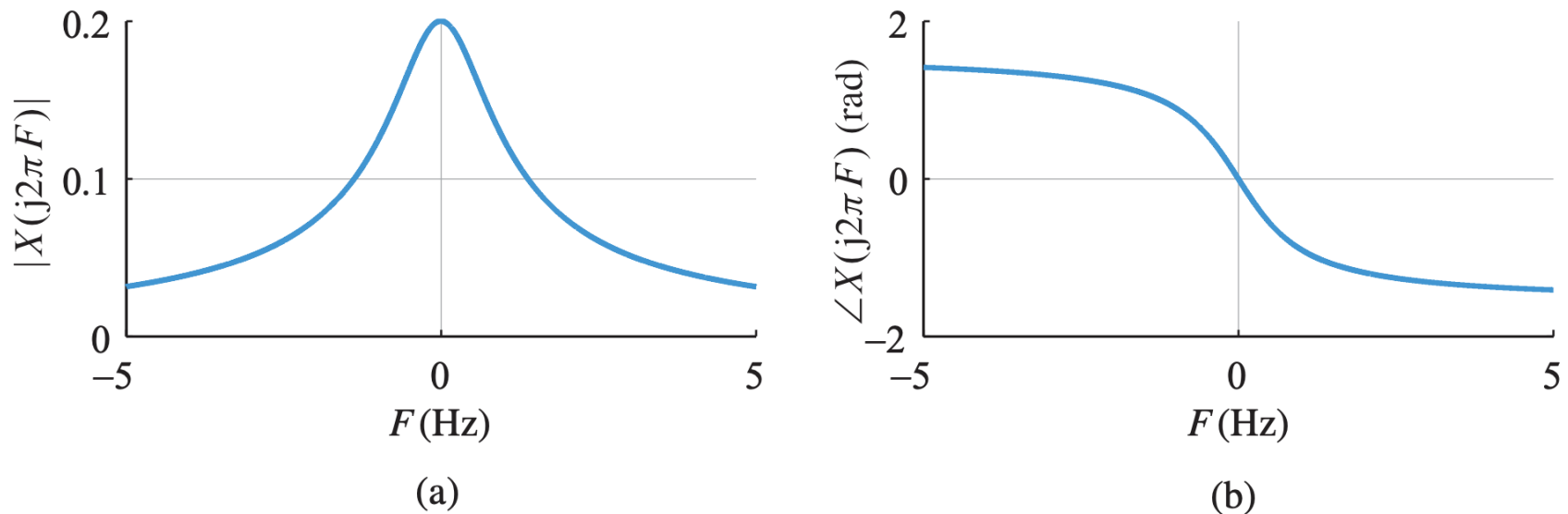


Figure 4.15 Fourier transform of the signal $x(t) = e^{-at}u(t)$ for $a = 5$. (a) Magnitude and (b) phase of $X(j2\pi F)$ in the finite interval $-5 < F < 5$ (Hz).

$x(t)$ is a real function of $t \rightarrow |X(j2\pi F)|$ has even symmetry, $\angle X(j2\pi F)$ has odd symmetry

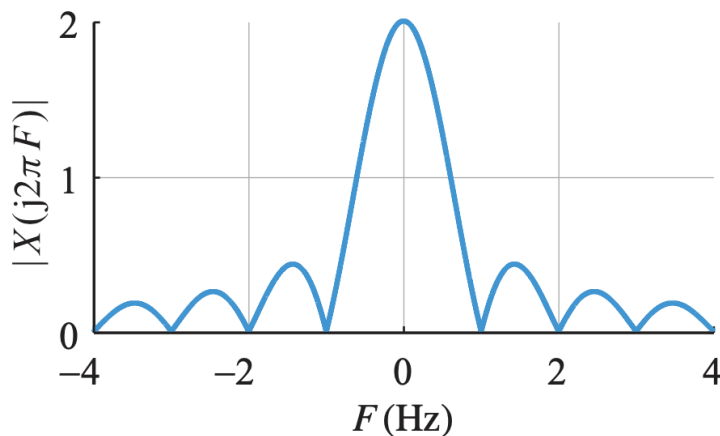
Fourier Transform for CT aperiodic signals

- Example 2: Spectrum of rectangular pulse signal

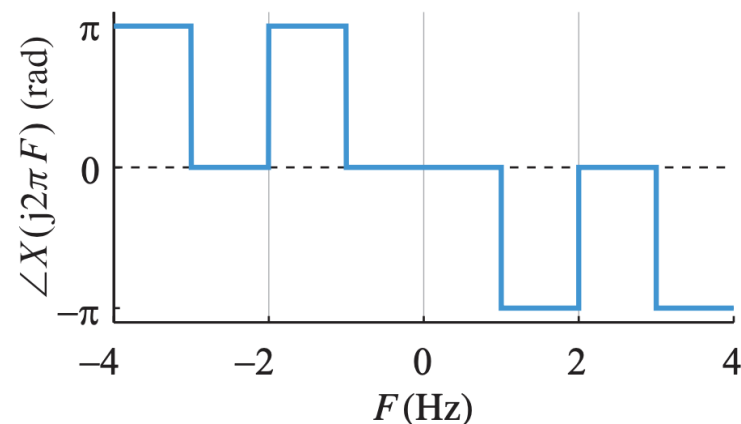
$$x(t) = \begin{cases} A, & |t| < \tau/2 \\ 0, & |t| > \tau/2 \end{cases}$$

This signal is absolutely integrable for any finite τ . Its spectrum is computed as

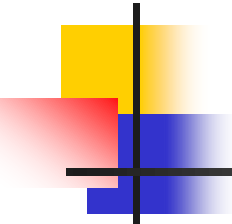
$$X(j2\pi F) = \int_{-\tau/2}^{\tau/2} A e^{-j2\pi Ft} dt = A\tau \frac{\sin(\pi F\tau)}{\pi F\tau}.$$



Magnitude spectrum ($A=2, \tau=1$)



Phase spectrum



Fourier series for DT periodic signals

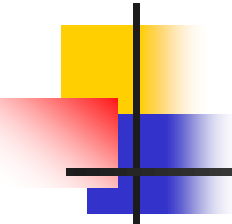
- Discrete-Time Fourier Series (DTFS) pair

Fourier Synthesis Equation

Fourier Analysis Equation

$$x[n] = \sum_{k=0}^{N-1} c_k e^{j\frac{2\pi}{N}kn} \quad \xleftrightarrow{\text{DTFS}} \quad c_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}.$$

- *Analysis equation*: analyzes (“breaks-up”) a periodic signal $x[n]$ into a set of N harmonic components $\{c_k \exp(jk\omega_0 n)\}$, where $\omega_0 = 2\pi/N$
- *Synthesis equation*: synthesizes the signal $x[n]$ from its N harmonic components
- N : fundamental period of periodic sequence $x[n]$



Fourier series for DT periodic signals

- Parseval's relation

The average power in one period of $x[n]$ can be expressed

$$P_{av} = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2 = \sum_{k=0}^{N-1} |c_k|^2.$$

- $|c_k|^2$: portion of the average power of $x[n]$ that is contributed by its k th harmonic component
- *Power spectrum*: The graph of $|c_k|^2$ as a function of $f = k/N$, $\omega = 2\pi k/N$, or simply k